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Be Aware of Your Text Messages: Fraud Attempts Identification Based on Semantic Sequential Learning for Financial Transactions through Mobile Services in Bangladesh

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Abstract

The growth of mobile financial services over the last decade is well accepted by the people of Bangladesh. However, although reported here and there, a complete understanding of illegal payments and fraudulent transactions is still missing. This paper studies one of the popular and experienced financial transaction services of Bangladesh, Bkash, to identify fraudulent text messages through the service. We propose an improved classification method for fraudulent messages through feature extraction and a hybrid deep learning (DL) approach with ML, which combines a convolutional neural network with sequential learning, referred to as the customized Bkash fraud semantic sequential learning (CBF-SSL) model. Seven machine-learning classifiers and five deep-learning sequential models are experimentally evaluated within the hybrid classifier. The performance of the fraud detection system is assessed using the loss function and confusion matrix on a dataset of 500 Bengali text messages. The performance evaluation demonstrates that the hybrid model of convolution neural network (CNN) using long short-term memory (LSTM) outperforms all other classifiers with the lowest error

rate of 0.2532%, the highest F1-measure of 75.00% with the highest test accuracy. The proposed text features in combination with CNN and LSTM prove to be highly effective in detecting fraudulent messages.

Keywords: Bengali text, Bkash services, convolution neural network, deep learning, fraudulent transactions, semantic sequential learning

1 Introduction

With the objective of facilitating rapid and convenient digital transactions, the population of Bangladesh has embraced the use of various mobile financial services (MFS) since 2011. Till now a number of MFSs are operating in different cohorts of the country including bKash, Rocket, Nagad, Upay etc [1]. Among those, the largest mobile financial services provider in Bangladesh, Bkash [2, 3], offers a range of financial services to customers, particularly those in rural areas by making financial transactions easier. These services include cash in, cash out, money transfer, business-related transactions, bill payments, remittance and so on. However, despite being the largest financial service provider, records show several fraud attempts in the Bkash platform [4].

Nowadays, short message services (SMS) targeting to dodge someone with a BKash account and stealing money saved in customers' BKash accounts is not uncommon [5]. Bangladesh is a small country with six different mobile operators (Grameen phone, Robi, Teletalk, Airtel, Citycell, and Banglalink)[6] with a huge number of subscribers. According to the BTRC (Bangladesh Telecommunication Regulatory Commission) report in 2020, 10M [7] messages are delivered daily. There are various methods of defrauding individuals through misleading messages, such as by presenting unexpected offers. Malicious SMS refers to any kind of illusion text message that is delivered from several fraud mobile networks [8]. Researchers discussed occurrences of fraudulent messages targeting mobile users, as outlined in [9], which provoke users through text messages. These messages, facilitated by malicious software, are responsible for siphoning significant amounts of cash from users' accounts, as described in [10] and [11]. A fraud finds the easiest way to communicate or transfer fake messages by using telecommunication systems with mobile phone users [12]. Numerous victims have been targeted through their smartphones by responding to fraudulent messages from attackers [13] [14] and finally sacrifice their money, PIN number, account details, etc. [9]. Therefore, identifying any fraudulent attempt through text messages or voice calls and searching for the real identity of those offenders is crucial for ensuring securing identities for financial services [15] [16].

This study identifies fraudulent messages by employing adaptive learning techniques on existing message datasets. Our challenge lies in the complexity of Bangla structure learning, where the same words can convey different meanings and pose significant syntactical challenges during text message detection. Consequently, word retrieval in a fraud context poses a significant obstacle for this study. We focused on the supervised machine learning approaches to identify message authentication.

This supervised approach outperformed by leveraging potential fraud word features mapped with the binarization technique. Therefore, we choose two classes fraud and non-fraud for Bangla messages according to authentic or fake word selection, where The filtration of malicious text messages on mobile phones has not been programmed yet, and users require an advanced method of message filtration.

In [17, 18], the authors discussed in their review work that cyber attacks may occur in various communication systems. Furthermore, the authors explore machine learning (ML) and deep learning as fields capable of detecting the true nature of cyber attacks, as well as the necessity for message or communication authentication. Our approach in identifying message authentication performance measures and loss optimization is evaluated using machine learning (ML) algorithms and deep neural networks (DNN) algorithms. To the best of our knowledge, no such fraud detection has been investigated yet considering the language 'Bangla' for text messages, which can be solved by Machine Learning and Deep Neural Network-based architecture. It is to be noted here that previously, supervised Learning methods, including LSTM (Long Short-Term Memory), CNN (Convolutional Neural Network), and BiLSTM (Bidirectional Long Short-Term Memory), have been used for different datasets comprising numerical, image, video, audio speech, and text in Bangla, except for fraud message identification [19, 20]. In deep neural networks, several layers are responsible for text information processing by leveraging all word weight input encoding [21][22]. In deep theory, we fine-tuned our model according to the input, and then we performed cross-attention between messages and corresponding class labels. One of the major components is the activation function, which works for optimizing the loss value. However, we found deep learning more suitable in classification, 3D tensor-based problems like NLP (Text messages) identification [23]. The binarization problem can be solved using machine learning, which covers a supervised approach to labeled data. Supervised approaches such as Logistic Regression, Decision Tree, Random Forest, Multinomial Naive Bayes, K-nearest Neighbor, Support Vector Machine, and Stochastic Gradient Boosting are applicable to classification problems where most of the Bangla language or different language identification is involved in it [24][25].

In the context of Bengali natural language processing (NLP), several approaches including deep learning (DL) techniques have proved their capabilities in several fields, such as handwritten Bengali text detection [26–30], using Deep learning Identification of Bangla overlaid text on videos [31–33], hand sign language to text using Neural Network [34], Automatic Speech Recognition [35–38], Machine Translation [39–41], Text Categorization [42–44] and Bangla Emotion Detection [44–47]. These are some works relevant to our methodology to propose a novel study on Bangladesh Bkash financial transaction awareness by detecting fraudulent texts using ML and DL. However, to the best of authors knowledge, no existing work focusing on text detection is available in Bengali dialect. We argue that detecting ambiguous behavior from text patterns is helpful in various sectors, including defense, banking transactions, or the health sector. Thus, our study is crucial for minimizing physical investigations, ensuring maximum service with minimal issues for bKash (and potentially other financial services in Bangladesh and similar contexts), and enhancing communication security. The novelty of our work is the Bengali language binarization technique developed that

can distinguish the authenticity of messages from fake ones. In addition, we provide an optimized technical solution that can prioritize the relevant words.

For this study, we focus on mobile texts, and short message services (SMSs), for further information analysis tool integration, which is available on users' mobile phones. People are connected through SMSs; therefore, we choose to collect all text messages from users' phones and use these text inputs for adaptive learning. We target malicious content where Bkash users lose their accounts and balance, aiming to develop awareness among Bkash customers. Therefore, we approached this technique, so people could identify the authenticity of messages.

To briefly highlight the implementation process, all SMS (texts) that come into a user's phone from outsiders are labeled by two categories: Fraud and Partial Fraud. After annotating fraud and non-fraud text messages, we propose a novel model named CBF-SSL (Customized Bkash Fraud Semantic Sequential Learning). A two-step approach is followed for the implementation. Firstly, we perform feature engineering, and secondly, we apply cross-attention focusing on input encoder embedding with an effective loss function. Collecting all the data inputs, we annotate the textual format after pre-processing input data. All the extracted weighted string data is converted into numerical data and then mapped with the semantic features. Finally, we engage in sequential learning by combining CNN, LSTM, and BiLSTM algorithms. Few researchers have worked on text mining using CNN and BiLSTM [48]. Nonetheless, we introduce a new model approach to text mining, with its performance evaluated by word sequence probability using the CBoW/bilingual (n-gram) method that captures word frequency and text similarity. We focus on two different models (CNN and LSTM) for generating a new architecture because LSTM [48] prediction accuracy is high, and it is capable of learning long sequences of text using short-term memory. We compare performance scores by referencing some state-of-the-art works related to this study. We show the supervised learning model's performance in terms of a few studies focused on ML for their dataset problem solution and performance purpose [25]. Figure 1 shows the architecture of the proposed workflow on malicious spam text detection. Overall, the main contributions of this paper are stated as follows:

- This study proposes a novel mechanism named the "customized Bkash fraud semantic sequential learning (CBF-SSL)" model, that can detect fraudulent text messages in an automated and adaptive manner.
- This study contributes to an efficient understanding of the Bkash fraud context by leveraging syntactical relationships.
- Due to adaptive learning, we focused on input embedding for over-extracted features containing fraud-suspect words. These words markedly differ from others in the messages. Consequently, the term ranking feature supports the extraction of priority-based features in fraud cases. Moreover, the proposed method demonstrated an acceptable performance, obtaining a 75% accuracy on Bengali text.

The remaining parts of the paper are organized, as follows: Section 2 presents relevant literature. Section 3 describes the methodology and the structure of the proposed approach. Section 4 delineates the experimental setup and performance analysis. We discuss the result findings in Section 5 and conclude the paper in Section 6.

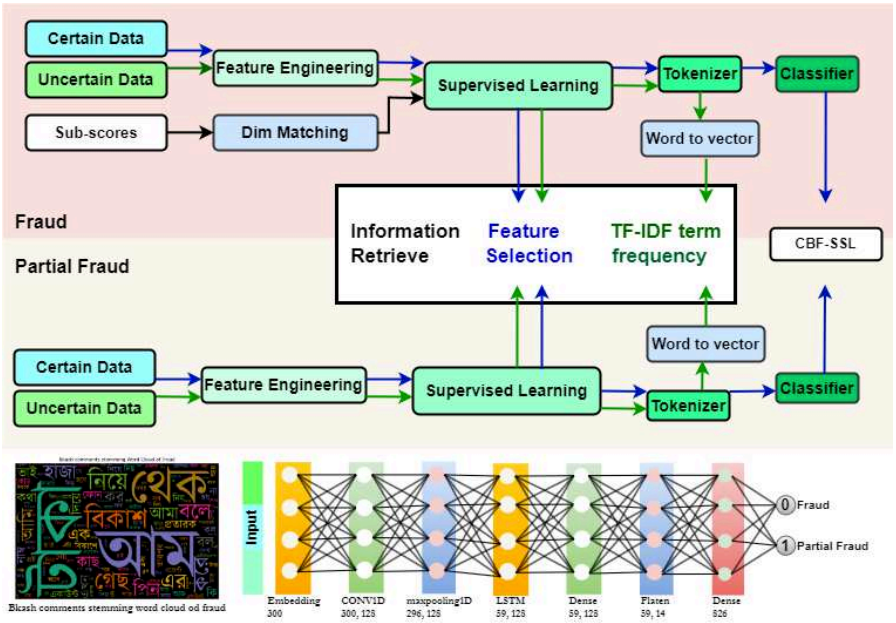


Fig. 1: Architecture of the proposed workflow on malicious spam text detection.

2 Literature Review

Although several cybersecurity awareness sessions have been practiced and proposed all over the countries [49], alarmingly, fraud attempts through messages are still increasing. Interestingly, there is no existing literature on detecting and discussing fraudulent BKash texts. This section illustrates relevant works and depicts our motivation for this study.

2.1 Survey-based studies on fraud awareness in financial transactions on Bkash

We examined several survey studies focusing on public opinion, as articulated by interns from various organizations. These studies collectively suggest that individuals are vulnerable to fraudulent activities. Moreover, in such blogs, posts, and online newspapers we found fraud cases are daily occurrences nowadays. The BRAC organization [50] is a local financial support system, their survey results demonstrate their research on occurrences of fraud attempts and present statistics on fraudulent transactions. All the studies indicate that Bangladeshi residents use Bkash accounts for fast transactions, moreover highlighting an enormous risk of fraud attempts as well. In [51], the authors discuss the necessity of utilizing financial transactions and the possibilities of Bkash, which have been creating new opportunities for quick transactions over mobile phones. They mention that users can fall into various traps by sharing personal information such as PINs or becoming victims of SMS scams. In [50], a study

on fraud risk management reveals several fraud cases that occurred in Bangladesh. It illustrates that MFS fraud can occur if people respond to fraudulent messages, and suggests that extensive data analysis may offer a solution, possibly through ML or DL. Thus, we focus on the Bengali NLP (BNLP) fraud attempt at complex structural analysis with a promising performance rate. In [8], the authors present empirical findings from a fraud case study where they propose developing data processing into a self-adaptive system. The study demonstrates forecasting results on fraud occurrences, enhancing safety and security in combating fraudulent calls and messages [15]. Additionally, [30] introduces a statistical analysis using ML techniques. Our focus lies on simple regression, regression coefficients, and variance across different features of fraud-related text data. To the best of our knowledge, there is a deficiency in adaptive learning with a cross-attention system for predicting Bkash fraud text over both ML and DL techniques.

2.2 Technical studies on machine learning and deep learning

In this study, we are discussing Bengali text, as there are few studies available in different languages, such as English, that address their financial methods by leveraging ML or DL focusing topic-based modeling. Developing a self-adaptive learner using Bangla text poses a few challenges in feature engineering. Therefore, there are a few technical studies we would like to discuss that have technical similarities with our system approach. In [52], crucial work was done proposing a methodology about malicious intent using 500 malicious email contents to detect fraud, later trained with BERT followed by a text summarization process, as well as constructing their own Transformer model. In [53], they collected 600 million tweet texts from spam and performed classification using several ML models, acquiring more than 90% accuracy. Another study revealed the last 20 years' statistical analysis of 5 large different datasets based on security, showing different statistical approaches. However, there are limitations in text analysis and positive and negative sentence correlation identification [54]. In [55], they proposed multi-level text identification on two different datasets containing both malicious and non-malicious comments and emails, experimented on LSTM, and gained 92.7%. However, their study was incapable of addressing Bkash Fraud. In [56], it was stated that managerial fraud comments detection using DL. Another study [57] from NLP used an ML algorithm that accurately identified 95%, which is a precision score on two different datasets containing over 10,000 individual emails. In [58], another study of ML comparative study using a classifier on different anti-phishing features. According to the study, 73,575 different URL texts were processed by the DT, Adaboost, K-star, RF, SMO, and NB for text identification. In [59], a novel approach of smishing (SMS and Phishing) features was evaluated by ML classifiers' algorithm and had two phases, among them one had eleven features and another one had four features of spam text, which achieved 96% accuracy on the Neural Network Classifier.

2.3 Technical studies on different languages

We reviewed various fraud detection studies conducted in different languages, and a few studies indicated that neural networks could handle the classification problem, especially for text learning. As our data is a primary dataset, we employed the feature engineering method by following the text extraction method through word2vector conversion. In some studies, we observed that texts in other languages were processed using their own selected engineering methods, which are not directly implementable in Bangla text. Therefore, we opted for BLTK for vectorization. Due to methodological differences, we followed some of these studies available in different languages in various ways. Based on this review, our aim is to propose a deep combined model (CBF-SSL) capable of efficiently detecting Bangla Bkash fraud compared to existing models. We selected promising models that demonstrated excellent accuracy in text processing while studying these ML and Deep Learning studies.

In [60], a hybrid model of CNN and LSTM were proposed using the Twitter dataset. Additionally, [61] demonstrated that their baseline architecture LSTM and BI-LSTM outperformed with 98.26% accuracy. The study also showcased several ML classifiers' accuracy on Arabic and English spam messages. SVM, MNB, DT, KNN, Adaboost, Extra trees, Logistic Regression, and Bagging Classifier were used for performance comparison on an SMS spam dataset collected from the UCI repository, comprising about 5574 English and 2730 Arabic texts, which is a secondary dataset. Another novel work focused on four different datasets, including IMDB, DBpedia ontology, 20 newsgroups, and Reuters-21578 dataset, developing an LSTM and RNN classification approach, with the model performance reaching more than 86% [62]. In [63], a study detecting malicious directories allowed the implementation of DL-LSTM and DL-RNN. A similar study on malicious sentiment text using CNN and LSTM [64] achieved a 0.83 AUC score using benchmark data. Turkish language [65] involved 800 spam emails using LSTM with higher accuracy and a low loss curve. In [66], a new study with simple malicious messages in the English language achieved a 90.33% score, while without the English language, it scored 96.40%. Moreover, they proposed a 'SmiDCA' (SMIshing Detection based on Correlation Algorithm) phishing model. This research is a kin to our approach, which demonstrated DL and ML implementation on fraud and non-fraud texts around 18366 labeled email messages. They separately processed 3416 phishing and 14950 real texts using SVM, DT, NB, LSTM, CNN, and Word Embedding as models. Their work focused on one input feature and one output feature. Word Embedding yielded an optimum output with higher accuracy of 98.69% [67], and SVM achieved 95.66% [68]. Additionally, other NLP work employed graph convolutional network (GCN) techniques for different language adaptation, such as Arabic and English, leveraging profound cross-attention techniques, which achieved a 98.2% accuracy [69]. Table 1 presents the comparative summary of related works.

In this paper, we would like to highlight the availability of extensive research on text mining utilizing machine learning and deep learning methods in various languages. These studies cover several aspects of text mining, similarity matching and text transcription; however, there is a notable gap in the research specifically addressing Bangla text in the context of Bkash fraud. Some studies have focused on smishing

detection, Twitter text, email content, and other relevant texts. Nonetheless, none of the studies have utilized Bkash financial transaction-related text sms as input mapped with semantic decisions. All these studies, which predominantly focused on languages such as English, Arabic, and Turkish, showcased cross-attention with adaptive learning, demonstrating performance. However, upon reviewing all the studies, we deduce that identifying Bkash fraud in Bangla text necessitates different feature engineering with a fine-tuned architecture capable of comprehending double-meaning sentences and detecting fraudulent messages.

3 Research Method

The proposed methodology is delineated across three distinct sections. Each section highlights the significance of dataset preparation in facilitating training and testing processes, as detailed in the initial section. In consideration of customer security, we anonymize the data during dataset preparation. This process encompasses data cleaning, validation of data templates, cleaning procedure steps, and the final selection of root words. Secondly, the study showcases the adaptability of Supervised Learning and Deep Learning in sections 4.2 and 4.3, respectively. In the supervised learning section, the study illustrates how machine learning models select the K best matches to optimize loss and yield effective results. Employing a supervised approach aids in decision-making and enhances the model structure, facilitating the detection of fraudulent text within various types of messages. Furthermore, in conjunction with embedding text through binarization in deep learning studies, we have developed sequential learning for text mining [70, 71], which proves to be more adept at memorizing sentences and achieving a high predictive score in test cases. This cybersecurity-based fraud attempt study underscores the importance of sequential learning. The proposed working process for identifying fraud or partial fraud cases is depicted in Figure 1.

3.1 Dataset preparation

In this section, we elucidate the steps involved in data collection, feature engineering methods, and all features extracted from the text. Regarding the collection process, we demonstrate how the data were gathered, and anonymized, and class labels were selected. Additionally, we have segmented the data processing and feature engineering methods into four distinct parts for a more comprehensive understanding such as (i) Data collection, (ii) Data labeling, (iii) Pre-processing steps, and (iv) Feature extraction methods.

3.1.1 Data collection

Initially, data is collected from Mendeley. The dataset is meticulously managed to ensure that it adheres to stringent privacy and security protocols, thereby guaranteeing that it does not breach any privacy or security measures. This dataset mainly contains Bkash scam-related data from public comments, on various types of sites, including Facebook, YouTube, and blogs, where individuals share their experiences

Table 1: Comparative summary of related works

Article ref.	Approach objective	Used methods	Dataset type
[52]	Identify malicious email	BERT model Accu.: High	Quantity: 500 emails self collected data repository: text summary from email
[53]	Classify SMS and identify spam messages	Different ML models Accu.: 90%	Quantity: 600M tweet spam messages self collected repository
[54]	Classify text messages	5 Different classifiers Accu.: High	Private Dataset
[55]	Multi level malicious identification of email text	LSTM Accu.: 92.7%	Quantity: Two different dataset English language smishing messages
[57]	Classify text and identify spam images	ML models Accu.: 95% Pre.	Quantity: 10000 email from 2 different dataset.English language Smishing text
[66]	Classify fishy text and identify spam messages	ML and DL models Accu.: English 90.33%, 96.4% non-english language	Quantity: 3416 + 14950 fishy and non fishy email texts from 2 different dataset. English language Smishing text
[67]	Classify fishy text and identify spam messages	SVM, DT, NB, LSTM, CNN and Word Embedding Accu.: English 98.69%	Quantity: 3416 + 14950 fishy and non fishy email texts from 2 different dataset. English language Smishing text
[58]	Identify spam messages and inspect included URL	ML: DT, Adaboost, K-star, RF, SMO, NB Accu.: Sufficient	Quantity: 73,575 Smishing text Private data collection repository
[59]	Classify SMS and identify malicious messages	ML models Accu.: 96%	Quantity: 11 features non-English dataset
[60]	Identify the smishing messages	CNN and LSTM	Twitter data from different sources
[61]	Identify SMS spam messages	LSTM and Bi-LSTM, ML classifiers Accu.: 98.26% ML classifiers	UCI machine learning repository: Arabic 2730 and English 5574 SMS dataset
[62]	Identify SMS spam messages	LSTM and RNN Accu.: 86% ML classifiers	Quantity: 21578 texts UCI ML repository: 4 dataset-IMDB, DBpedia, 20 newsgroups, Reuters
[65]	Identify SMS spam messages	LSTM Model Accu.: high and low loss	Quantity: 800 spam messages Turkish language dataset

through adaptive learning. These individuals shared authentic text messages they received on their mobile phones, copying and posting them on different platforms in an attempt to reclaim lost funds. Consequently, our primary dataset creation adheres to ethical standards, as it was anonymized during the process. We utilized information from Bkash's official Facebook page and Instagram, reliable sources where users share their fraudulent experiences, for adaptive learning. For data storage, we opted for a cloud-based system and organized the fraud cases in a structured learning format.

Two key features were employed: the first attribute encompassed all messages gathered from diverse sources, while the second provided data labels, categorized as Fraud and Non-Fraud. Initially, 50% of the data was manually labeled, while the remainder was labeled using unsupervised learning techniques. We collected a total of 500 data points from the aforementioned platforms.

3.1.2 Data labeling

Utilizing adaptive learning data labeling is an essential step in addressing classification-based problems. Initially, targeted data is manually labeled to comprehend features for training purposes. The dataset, post-collection, is categorized into two distinct groups: fraud and partial fraud. A fraud label is utilized when an imposter attempts to deceive individuals by soliciting their PIN or OTP numbers to illicitly access and withdraw funds from their Bkash accounts, or when unauthorized deductions or theft occur without valid authorization codes. Partial fraud indicates instances where imposters attempted but failed to execute fraudulent actions, or messages resembling Bkash advertisements or customer inquiries, which we included in our dataset. Upon labeling our dataset, it assumes the following format as depicted in Table 2.

Table 2: The sample dataset of Bengali fraud text is categorized based on a binary class label. Fraudulent data is identified by distinguishing users' fraudulent messages from partial fraud cases, with the latter comprising all other messages including those sent by Bkash.

Comments	Label
আমাকে এই পর্যন্ত ১০০ বার ফোন দিয়েছে এই প্রতারক চক্র কিন্তু সহজ কোন মাধ্যমে জানানোর ব্যবস্থা না থাকায় আমি কাউকে জানাতে পারিনি। (I have been called 100 times by this fraudster but I could not inform anyone as there is no easy way to inform.)	Partial Fraud
আমার অ্যাকাউন্ট থেকে পাঁচ হাজার টাকা নিয়ে গেছে একটা নাম্বার থেকে ফোন দিছিল আমিও বুঝতে পারি নাই যা যা বলছে করছি নিজের ভুলে সব গেল টাকাটা খুব প্রয়োজন ছিল । (Five thousand taka were taken from my account. Someone called from a number. I didn't understand what I was saying. I forgot everything. The money was very necessary.)	Fraud
বৃত্তির কথা বলে আমার থেকে আজকে ৩৫০০০ হাজার টাকা নিয়ে গেছে আল্লাহ যাদের ক্ষমা না করুক। (35000 thousand taka have been taken from me today by talking about scholarships, may Allah not forgive them.)	Fraud
প্রতি সপ্তাহে আমার বিকাশ একাউন্ট থেকে ১৫ টাকা করে কেটে নেওয়া হচ্ছে। (15 taka is being deducted from my bkash account every week.)	Partial Fraud

3.1.3 Feature engineering method over Bkash fraud Bengali texts

In order to remove irrelevant parts or elements from the text dataset, preprocessing is a crucial step. This process generates clean text, which is essential for creating a root word tokenizer for subsequent processing techniques. Various technical preprocessing tools are employed to collect extracted features. The steps we followed are briefly outlined below and depicted in Figure 2, illustrating all features before generating an effective Bengali tokenizer.

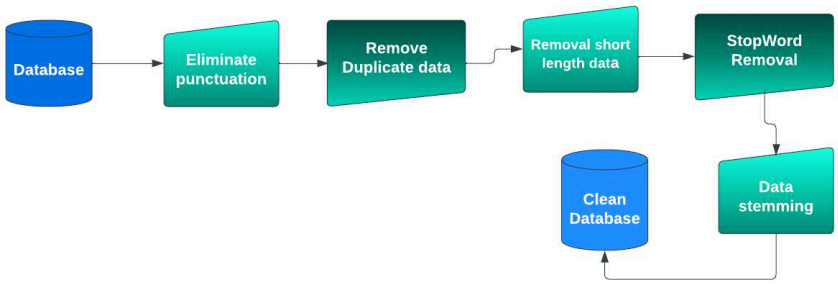


Fig. 2: Data preprocessing steps on fraud and partial text.

- *Removal of Duplicate Data:* Duplicate text data is eliminated to produce a unique and new collection of instances, enabling the model to predict fraud cases more accurately. The presence of duplicated dataset instances can introduce biases into the model. In Python, the presence of duplicate data in a dataset can be detected using the duplicate() function. Additionally, we implemented the following preprocessing steps:
- *Removal of Short Length of Data:* We analyzed the dataset and removed data instances with lengths less than three. Text with short lengths often lacks sufficient information about fraud attempts. For instance, a single-word comment such as 'advertising' or a mention of 'PIN request' does not provide informative content. Therefore, eliminating short sequences of text data allows the model to focus on longer, more informative sequences. Figure 3 illustrates the frequency distribution of data lengths, starting from a minimum of 4 and reaching a frequency level between 15 to 20.

- *Maximum Length of a comment:* 123 *Minimum Length of a comment:* 4
- *Remove Punctuation:* Bangla sentences often contain various symbols and words, including colon (:), comma (,), full stop (.), exclamation mark (!), and irrelevant Bangla words. Before implementing the targeted data, it should only include words that carry meaning or have a relationship with the sentences. For the cleaning step, we utilized the Keras library for feature extraction with K value, tokenization, sequencing, and encoding.

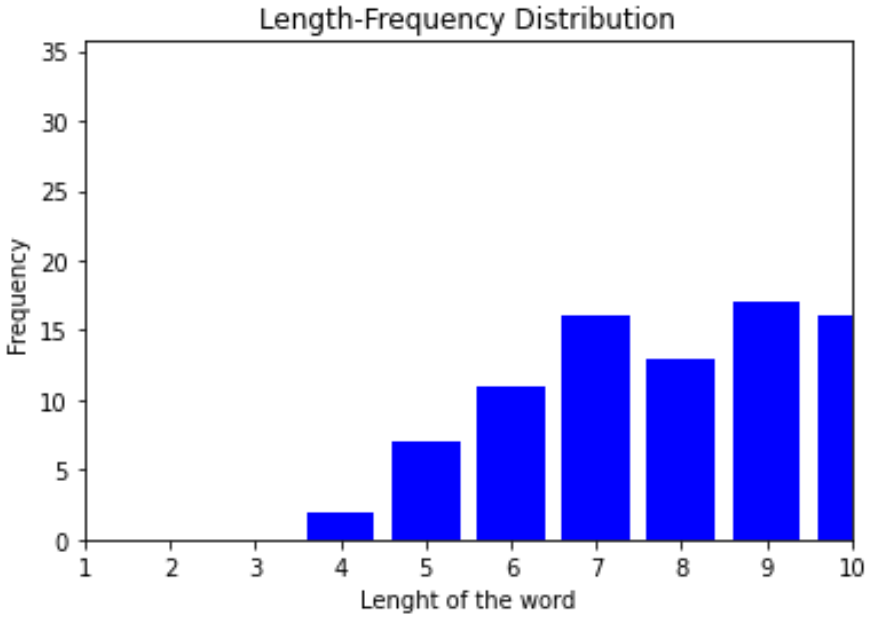


Fig. 3: Illustration of data length frequency distribution.

- *Word to Vector:* Vectorization is a crucial phase in the feature extraction process of machine learning. It involves translating text into numerical vectors, aiming to extract distinctive characteristics from the text for the model's training. To determine word weights, we employed a numerical statistic known as Term Frequency-Inverse Document Frequency (TF-IDF). Based on these weights, we created a tokenizer using the term ranking method. Words with higher frequencies will carry more weight, as identified by the term ranking process, and will be converted into vectors. Table 3 shows a bilingual illustration of word vectorization data.

- *Tokenization:* The Keras tokenizer facilitates the conversion of text into a machine-readable format. During this process, the raw text is segmented into tokens, comprising individual words and phrases. Tokenization involves analyzing the syntax to decode the meaning of the text. The numbering of each word is utilized to establish the vocabulary, with a maximum limit set at 5000 words.
- *Stopword Removal:* The process of removing Bangla stop words plays a crucial role in text preprocessing for natural language processing tasks. Stop words are common words in a language that do not carry significant meaning and are often excluded from analysis to improve the efficiency and accuracy of text-processing algorithms. In Bangla, stop words include frequently occurring words such as `এবং' (and), `কিন্তু' (but), `যে' (that), etc. The removal of these stop words is essential as they can skew the results of text analysis by adding noise to the data. Typically, a predefined list of Bangla stop words is used, and during the preprocessing phase, each word in the text is compared against this list. If a match is found, the word is eliminated from

Table 3: Bilingual illustration of word vectorization data

Comment	Unigram	Bigram	Trigram
জব বলে আমার থেকে ১২০০ টাকা এর কথা। ৫০০ টাকা ইনভেস্ট এর কথা বলে ১২০০ টাকা নিলো প্রতারক (Job says about 1200 taka from me. Fraudster took 1200 taka by talking about 500 taka investment)	[('আম', 1032), ('কর', 409), ('বল', 212), ('এক', 201), ('রত', 149), ('উন', 113), ('এর', 100), ('এখন', 92), ('এই', 87), ('রক', 84), ('পর', 75), ('হয়', 66), ('আর', 63), ('কথ', 60), ('কল', 60), ('আগ', 56), ('রপর', 53), ('আজ', 51), ('জন', 50), ('ইয়', 46)]	[('আম এক', 106), ('এক উন', 88), ('রত রক', 74), ('কর আম', 74), ('বল আম', 49), ('আম কর', 49), ('কথ বল', 47), ('আম আম', 45), ('আম বল', 31), ('এখন আম', 28), ('রত রণ', 23), ('রপর আম', 22), ('আগ আম', 21), ('আজ আম', 20), (' ', 19), ('ইয় আম', 19), ('বন কর', 18), ('আম রত', 18), ('আম কল', 18), ('আম আজ', 17)]	[('আম এক উন', 69), ('কথ বল আম', 23), ('এক রত রক', 17), ('রত রক আম', 11), ('এক উন কর', 11), ('এক উন বন', 10), ('কর আম আম', 10), ('রত রণ কর', 9), ('এর কথ বল', 8), ('আম বল আম', 8), ('আম কর আম', 8), ('আম নম বর', 8), ('এক উন লক', 7), ('রত রণ হয়', 7), ('আম মন কর', 7), ('কর আম মন', 7), ('কর রপর আম', 6), ('উন বন হয়', 6), ('উন লক কর', 6), ('এই রত রক', 6)]

the text. By removing Bengali stop words, the resulting text data becomes more concise and focused, facilitating more accurate analysis and interpretation.

In Table 4, we demonstrated a list of stop words that are necessary to remove for model output.

Table 4: List of Stop Word on Fraud and Non-fraud text data

অতএব (Therefore)	অথচ (While)	অথবা (Or)	অনুযায়ী (according to)	অনেক (many)	অন্তত (at least)	অন্য (other)	অবধি (until)	অবশ্য (of course)	আগামী (future)
উনি (they)	এঁরা (they)	একই (same)	একবার (once)	এখন (now)	কারণ (because)	তারপর (then)	তারা (they)	দেওয়া (give)	দেখতে (see)

- *Stemming:* Stemming aimed at reducing words to their root form or stem. Stemming helps in standardizing word variations, thereby improving the efficiency of text analysis algorithms. In Bangla, words often undergo morphological changes based on tense, gender, and other grammatical factors. Stemming algorithms identify and remove affixes from words to extract their base form. For example, the word `খেলা' (game) can be stemmed from its root form `খেল'. By applying stemming, variations of the same word are mapped to a single representation, reducing the vocabulary size and enhancing the accuracy of text processing tasks such as document classification, information retrieval, and sentiment analysis in Bengali text. Table 5 presents the sample dataset on fraud Bengali and English text messages after preprocessing.

Table 5: Sample dataset on fraud Bengali & English text messages after preproccessing

Comment	Clean Comments
আমার ৬০০০ টাকা হেক করেছে কাল ফন নাম্বার ব্লক করে দিচে (They hacked my 6000 taka yesterday,, and blocked my phone number)	আমার ৬০০০ টাকা হ্যাক করেছে, কাল ফোন নাম্বার ব্লক করে দিয়েছে (They hacked my 6000 taka yesterday blocked my phone number)
এটা প্রথম না আমি বিকাশ এজেন্ট প্রতারিত হয়েছি ২০ সালে তিন টাকা কম ৪০,০০০ টাকা নিয়েছে এই চক্র আপনাদের কাছে প্রথম হতে পারে আমাদের কাছে অনেক পুরানো| (This is not the first time I have been cheated by a Bkash agent 20 years 3 taka less 40,000/= taken this cycle may be first for you very old for us.)	এটা প্রথম না আমি বিকাশ এজেন্ট প্রতারিত হয়েছি ২০ সালে তিন টাকা কম ৪০০০০ টাকা নিয়েছে এই চক্র আপনাদের কাছে প্রথম হতে পারে আমাদের কাছে অনেক পুরানো (This is not the first time I have been cheated by a Bkash agent 20 years 3 taka less 40000 taken this cycle may be first for you very old for us.)

As an example, for visualizing root words, we generate a word cloud using all stemmed words, as depicted in Figure 4. Although the individual sentences may lack proper meaning, they all represent root words. Utilizing root words can optimize model output by reducing processing time when encountering similar root words.

- *Label Encoding:* Machines only comprehend values represented by 0 and 1; thus, the encoding process is used to convert target labels into numerical form (0, 1). Since our dataset is structured (comprising human-generated data), we initially labeled it manually in string format and subsequently converted these labels into



Fig. 4: Word cloud illustration for stemming data for fraud and partial fraud

binary format to reduce time complexity while requiring a larger storage space. We incorporated two features, one serving as input and the other as output. The output column contains a binary string category, necessitating encoding due to its binary nature, with only two target values - Fraud and Partial Fraud. Encoding is a crucial aspect of feature engineering for both supervised and cross-attention-based methods. It translates instances of fraud and partial fraud into 0 and 1, respectively.

- *Splitting of Dataset:* The split() function divides the dataset into training and testing sets. The training set is utilized for model training, including ensemble learning techniques. Conversely, the testing set is employed to assess the accuracy of model predictions. To feed the model effectively, we divided the dataset into three sections: training, testing, and validation. Eighty percent of the data is allocated for training and validation purposes, while the remaining 20% is dedicated to testing the model's performance.

Table 6: Classifier algorithm and its abbreviation

Classifier algorithms	Abbreviation
Logistic Regression	LR
Decision Tree	DT
Random Forest	RF
Multinomial Naive Bayes	Multi.NB
K-Nearest Neighbor	KNN
Radial Basis Function-Support Vector Machine	RBF-SVM
Stochastic Gradient Descent	SGD

3.2 Supervised machine learning approaches

We include 7 ML classifiers to solve our problems with a good outcome. Table 6 shows the classifier algorithm and its abbreviation. Supervised learning is dedicated to identifying targeted value on fraud text input [72] which is described below.

Forecasting any messages before an incident occurs with the customer, logistic regression predicts whether a message is likely to be true or false. Our analysis aims to determine whether the text suggests fraud or partial fraud. Unlike linear regression, which fits a line to all training data, logistic regression fits an S-shaped logistic function known as the Sigmoid Function. The sigmoid function ensures that the input's range falls between 0 and 1 [73]. A decision tree, known as a classification tree, categorizes items into categories by repeatedly splitting into decision nodes until reaching the leaf node, ultimately making a fraud or partial fraud decision based on the statement. For our dataset, the leaf nodes represent Fraud and Partial Fraud, with roots defined as input features [72]. Random Forest generates multiple decision trees and aggregates their predictions to obtain a more accurate and reliable prediction, avoiding overfitting, reducing training time for large datasets, and providing highly accurate predictions when dealing with vast amounts of data [74]. Using the Bayes theorem, computers predict labels for text using probability theory, which we employed to enhance prediction accuracy for Bangla text in our dataset [74]. KNN organizes new instances based on a similarity metric, calculating the distance between provided points and considering outlier points to distinguish its performance, utilizing the Euclidean function for detecting fraud and partial class labels. Point classification is determined by the categorization of its neighbors' values after distance calculation, value sorting, and voting values of neighbors [75]. The RBF kernel is a popular kernel function in machine learning, utilized in support vector machine classification to calculate the similarity or degree of proximity between two points, X1 and X2 [76]. SGD (Stochastic Gradient Descent Classifier) considers only one random row from available data, tailoring it to its attributes, thus reducing computational complexity and speeding up model learning as an optimization algorithm [77].

3.3 DNN architectural layers and models setup

3.3.1 Long-short-term memory

The working process of LSTM for two output classes, Fraud and Partial Fraud, involves a series of sequential steps in recurrent neural networks (RNNs). LSTM networks are particularly adept at handling sequential data due to their ability to retain long-term dependencies. In the case of fraud detection, LSTM models are trained on sequences of text data representing messages or transactions. These sequences are fed into the LSTM network, which processes them step by step, updating its internal state based on the information it receives at each time step. The LSTM network learns to distinguish between fraudulent and partially fraudulent instances by analyzing patterns and relationships within the text data. It does so by adjusting the weights of its connections during the training process, optimizing its ability to classify sequences accurately. Once trained, the LSTM model can predict whether a given sequence corresponds to fraudulent or partially fraudulent activity with a high degree of accuracy, making it a powerful tool for detecting fraudulent behavior in textual data. For the LSTM model implementation, we used 64 embedding dimensions and 32 batch sizes.

Epochs are the number of iterations that a training dataset makes around an algorithm; we used 10 epochs. Figure 5 illustrates the LSTM architecture with the hidden layer.

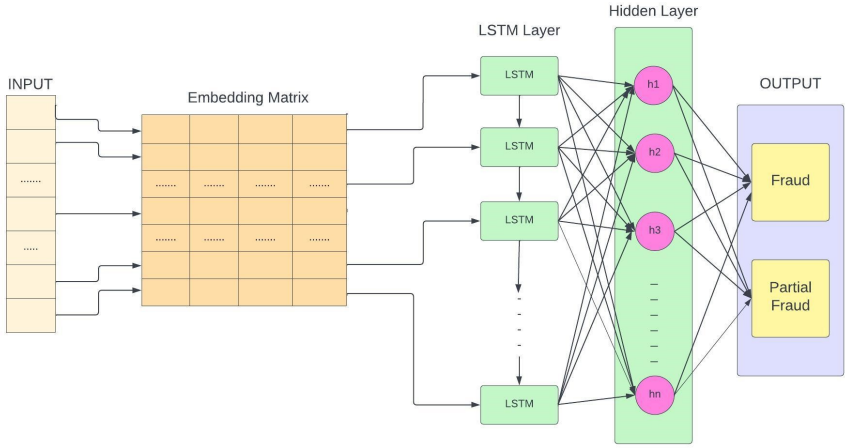


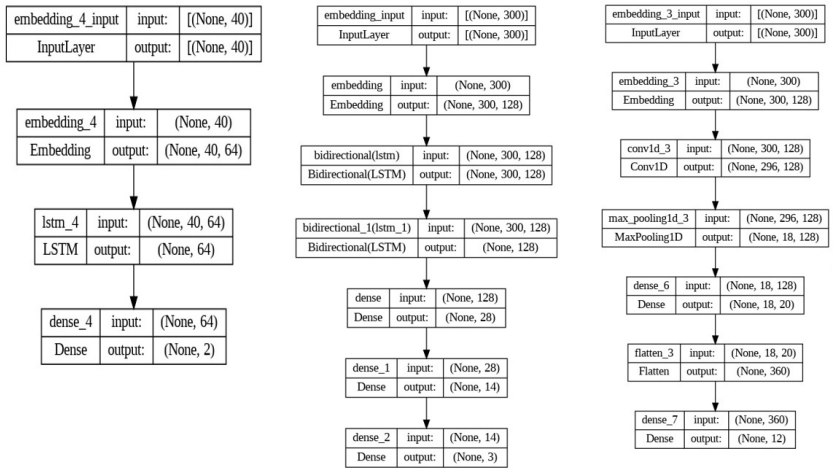
Fig. 5: Illustration of LSTM architecture.

3.3.2 Bidirectional LSTM (BiLSTM)

BiLSTM networks are a variant of recurrent neural networks (RNNs) that incorporate information from both past and future time steps when making predictions. In the context of fraud detection with two output classes, Fraud and Partial Fraud, BiLSTM networks offer a unique advantage by considering the context of the entire sequence of text data. Unlike traditional LSTM networks, which only process data in one direction (from past to future), BiLSTM networks process data in both forward and backward directions simultaneously. This allows them to capture dependencies and patterns in the text data from both directions, enhancing their ability to understand the context and meaning of the input sequence. In the case of fraud detection, BiLSTM networks can effectively analyze sequences of text data to distinguish between fraudulent and partially fraudulent instances. By leveraging information from both past and future time steps, BiLSTM networks can make more informed predictions about the nature of each sequence, ultimately improving the accuracy and reliability of fraud detection systems. The model is demonstrated in Figure 6 we employed 128 embedding dimensions, 64 batch sizes, and 10 epochs to build the bi-LSTM model.

3.3.3 Convolution neural network

CNNs are primarily known for their effectiveness in image recognition tasks, but they can also be applied to sequential data such as text for tasks like fraud detection. In the context of two output classes, fraud and partial fraud, CNNs can be employed



(a) LSTM based model (b) BiLSTM based model (c) CNN based model

Fig. 6: Deep learning embedding encoding-decoding with Maxpooling & dense layer full architectures.

to analyze sequential text data by treating each word or character as a feature map. The CNN architecture consists of convolutional layers that slide a set of filters across the input text, extracting local features at different positions. Pooling layers are then used to reduce the dimensionality of the extracted features while preserving their important characteristics. By iteratively applying convolution and pooling operations, CNN learns hierarchical representations of the text data, capturing both low-level and high-level features. Finally, fully connected layers are used to map the learned features to the output classes, Fraud and Partial Fraud. CNNs excel at capturing local patterns and dependencies within sequential data, making them a powerful tool for detecting fraudulent activities in textual data. The word embedding matrix served as the input for a 1D convolutional layer, which applied filters of size 128 to extract various patterns in the fraudulent text. This process generated a feature map by combining information from all sentences containing illegal content. Subsequently, a max-pooling layer was applied to each feature map to capture the most salient features directly related to fraud attempts. The output of the max-pooling layer was then flattened and passed to a fully connected layer to consolidate the weighted text output. This fully connected layer's output was utilized to classify fraudulent text messages. Our CNN model was constructed with 128 embedding dimensions, a batch size of 64, and trained for 10 epochs to ensure balanced output.

3.3.4 An ensemble model: CNN-BiLSTM

The Combined Model, which utilizes a CNN-BiLSTM architecture, integrates both CNN and bidirectional long short-term memory (BiLSTM) layers. Initially, CNN techniques are employed to generate word-level attributes, leveraging convolution and

max-pooling layers to create unique feature maps from per-character features such as word embeddings and character types. Following a word embedding process that represents each word as a dense vector, the data is fed into a 1D convolutional layer, applying multiple filters to extract local features and identify significant patterns within the text. This layer is capable of capturing essential n-gram features indicative of fraudulent content. The output from the convolutional layer is then inputted into a bidirectional LSTM layer, which processes the sequence in both forward and backward directions concurrently, enabling comprehensive capture of sequential dependencies and contextual information. The BiLSTM layer effectively retains temporal nuances associated with fraud patterns. The CNN-BiLSTM model is configured with 64 batch sizes, 10 epochs, and 128 embedding dimensions. Figure 7 illustrates the ensemble learning of malicious identification techniques.

3.4 Proposed method: CBF-SSL

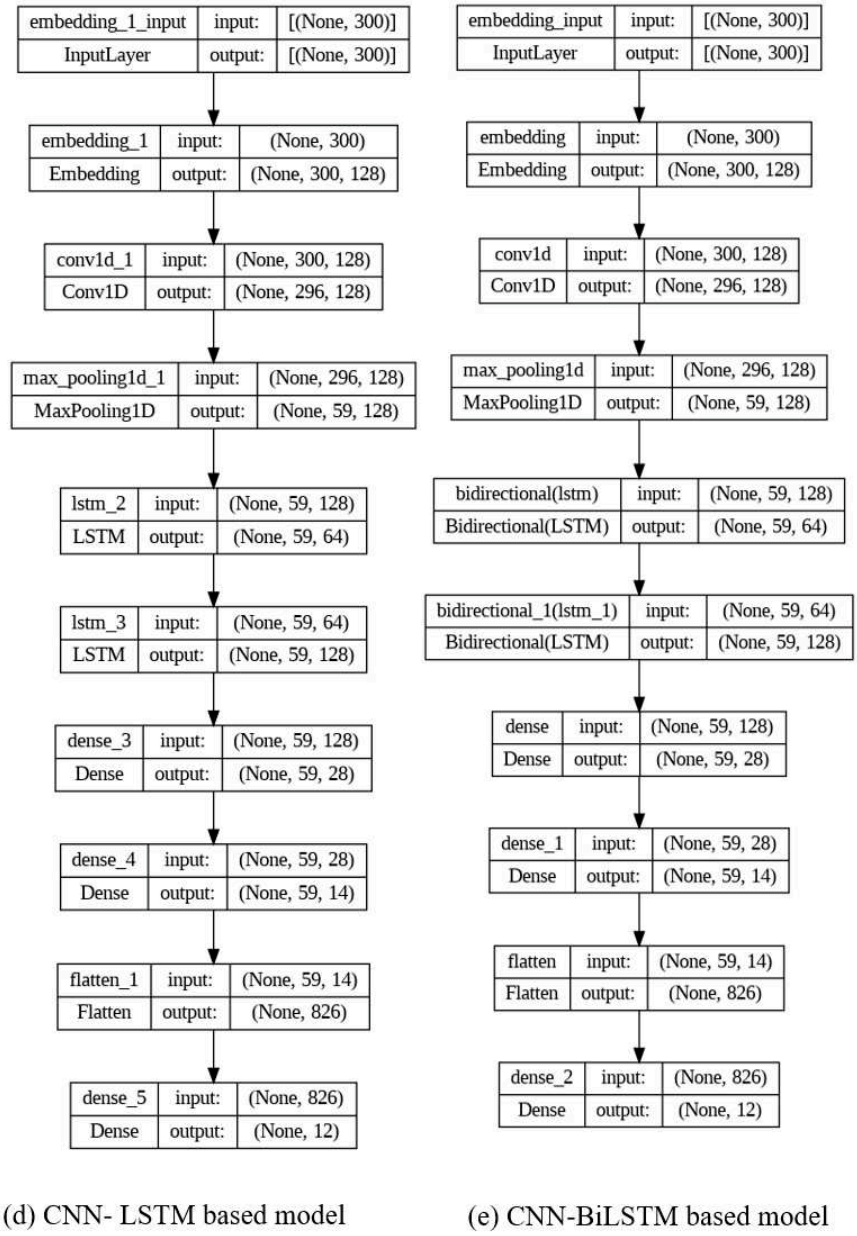
The ensemble model, which integrates CNN and LSTM yielded optimal results. This architecture comprises a CNN layer in the encoder and an LSTM layer followed by a dense layer outwardly. Here, LSTM decodes temporal features while CNN extracts features. The model benefits from its ability to handle large input structures, with CNN extracting desired features from sequences and LSTM assembling and decoding them. In this model, as depicted in Figure 7, input text data is first transformed into word embeddings. These embeddings are then inputted into a 1D convolutional layer, which applies a filter size of 128 to extract local features. Subsequently, a max pooling layer downsamples the feature maps to capture salient information. The output is passed to an LSTM layer, capturing sequential dependencies and modeling long-term dependencies in the text data. By utilizing LSTM, the model can understand temporal patterns and contextual information in fraudulent text. Fully connected layers are added for classification, further processing features, and making predictions about fraudulence. The output layer employs a softmax activation function to produce a probability distribution over classes (fraud or partial fraud).

3.5 Loss function

The loss function quantifies the disparity between the predicted output and the desired output. Throughout model training, the loss is computed in each epoch, as the biases of the model fluctuate to enhance accuracy while minimizing the loss rate, resulting in frequently changing loss values. In our model, we employed seven distinct loss functions: mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), R2 Score, variance score (VARScore), mean squared logarithmic error (MSLE), root mean squared logarithmic error (RMSLE), and mean absolute percentage error (MAPE), to compare the error rates. Table 7 shows the loss function and its abbreviation.

3.6 Optimizer

A model optimizer updates the model in relation to the loss function's output. To reduce error, we used the "Adam" optimizer on our model approach. Adam's complete



(d) CNN- LSTM based model (e) CNN-BiLSTM based model

Fig. 7: Deep learning embedding encoding-decoding with Maxpooling and dense layer full architectures.

Table 7: Loss function and its abbreviation

Loss function	Abbreviation
Mean Absolute Error	MAE
Mean Squared Error	MSE
Root Mean Squared Error	RMSE
R-Squared	R2 score
Variance score	VARScore
Mean Squared Logarithmic Error	MSLE
Root Mean Squared Logarithmic Error	RMSLE

name is adaptive moment estimation. It is a combination of momentum and rmsprop. Adam optimizer is suitable for textual datasets where the enormous length of text has a large weight and the optimizer needs low memory where computationally efficient too. In NLP tasks, such as sentiment analysis or text classification, the Adam optimizer helps in navigating through the complex parameter space of the neural network to find the optimal set of weights that can accurately capture the underlying patterns and relationships in the text data. By adaptively adjusting the learning rates for each parameter based on the first and second moments of the gradients, Adam ensures faster convergence and better generalization performance, making it particularly suitable for training deep neural networks on large-scale text datasets.

3.7 Confusion matrix

For evaluation purposes, we prepared a confusion matrix in Fig. 8 on Accuracy, precision, recall, and f1-score to check how many messages the model can correctly identify or not. The confusion matrix report evaluated performances by filtering all classified fraud texts that are truly positive in identification and all non-fraud texts that are partially fraudulent were identified by the true negative.

Accuracy: The percentage of total correct predictions is known as accuracy. It can also be described as a model's correctness in machine learning.

$$Accuracy = [\frac{(TP + TN)}{(TP + FN + FP + TN)}]X100\% \tag{1}$$

Precision: The percentage of accurate positive forecasts is known as precision. Precision aims to reduce errors made when assuming positive labels.

$$Precision = [\frac{(TP)}{(TP + FP)}]X100\% \tag{2}$$

Recall: Recall is a measurement used to determine true positives (TP). Recall also informs us of any inaccurate assumptions made by the model or cases that the algorithm misclassified.

$$Recall = [\frac{(TP)}{(TP + FN)}]X100\% \tag{3}$$

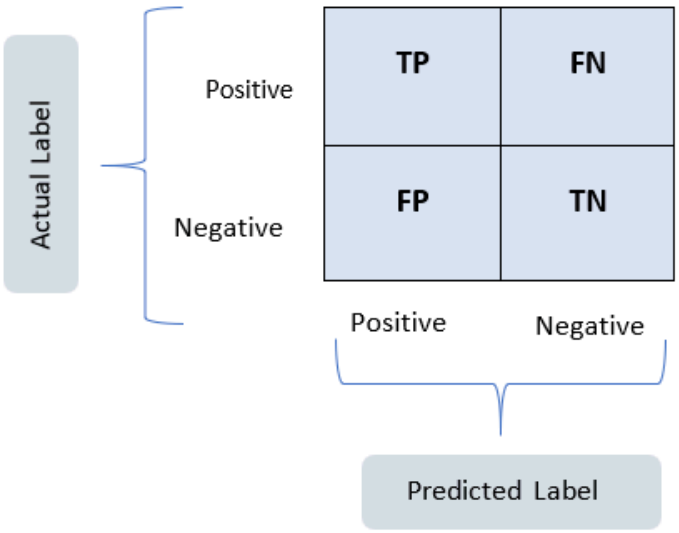


Fig. 8: Illustration of confusion matrix.

F1 Score: F1 score is a combination of precision and recall as it combines them in harmonic mean.

$$F1 - Score = 2X[\frac{(PrecisionXRecall)}{(Precision + Recall)}]X100\% \tag{4}$$

4 Experimental Setup and Performance Analysis

4.1 Experimental setup

For the hardware setup, we utilized a Core i7 laptop with 8.00 GB of RAM and a 64-bit operating system. As for the software setup, we employed the Jupyter Notebook. Initially, we mounted the file using Google Drive to facilitate our work and imported all necessary libraries and classes such as pandas, NumPy, TensorFlow, Keras, matplotlib, and sklearn. Google Drive served as the conduit for connecting with cloud-based data.

4.1.1 Model configurations of deep learning techniques

In the realm of deep learning, we applied three neural network models along with two ensemble learning approaches - LSTM, BiLSTM, CNN, CBF-SSL, and CNN-BiLSTM. Here, we outlined the experimental model configuration list for deep neural networks.

Table 8 illustrates all the layer descriptions that contribute to the outcomes in Bkash fraud analysis.

Table 8: Model details of the experimental conditions for DNN models

Models	Embed. layer	Conv1D layer	MaxPooling layer	LSTM layer	BiLSTM layer	Dropout layer	Classification Layer
LSTM	64	N/A	N/A	1 layer with 64 unit	N/A	1 layer with 20%	1 Softmax unit
BiLSTM	128	N/A	N/A	N/A	2 layer with 64 unit	1 layer with 20%	1 Softmax unit
CNN	128	1 filter layer, 128 filter, Filter size 5	1 layer with 16 pooling size	N/A	N/A	1 layer with 20%	1 Softmax unit
CNN-LSTM	128	1 filter layer, 128 filter, Filter size 5	1 layer with 5 pooling size	2 layer L1: 64, L2: 128	N/A	1 layer with 20%	1 Softmax unit
CNN-BiLSTM	128	1 filter layer, 128 filter, Filter size 5	1 layer with 5 pooling size	N/A	2 layer L1: 64, L2: 128	1 layer with 20%	1 Softmax unit

4.2 Performance analysis

4.2.1 Performance evaluation of supervised machine learning classifiers

To analyze the obtained Bkash scam comments or messages column and determine binary decisions, we have implemented several supervised models, including decision trees, random forests, multinomial naive Bayes, KNN, RBF SVM, logistic regression, and SGD.

Table 9 presents the accuracy, recall, precision, and F1 scores of the machine learning models for our input text, achieving a maximum accuracy of 77.88%. SGD outperformed every other algorithm; however, its test score was lower compared to our proposed model. Additionally, this section discusses the visualization of the confusion matrix using machine learning classifier models for Bkash scam-related output features.

Table 9: Performance score of ML algorithm for Bengali Bkash dataset

Models	Accuracy	Precision	Recall	F1-Score
LR	76.92	88.64	67.24	76.47
DT	64.42	67.80	68.97	68.38
RF	74.04	81.63	68.97	74.77
Multi. NB	74.04	84.44	65.52	73.79
KNN	69.23	80.95	58.62	68.00
RBF SVM	57.69	88.89	27.59	42.11
SGD	77.88	87.23	70.69	78.10

4.2.2 Performance evaluation of deep learning models

We employed two hybrid deep learning models, CNN-BiLSTM and CBF-SSL (Customized BKash Fraud Semantic Sequential Learning), in addition to three deep learning models, LSTM, BiLSTM, and CNN, in our investigation. This section describes the results of applying these models to our dataset. The maximum accuracy of our proposed model, which achieved the highest level of fraud messages, was demonstrated by Table 10. With Bangla text data, CNN-BiLSTM yielded 75% accuracy, making it a viable suggestion provider for Bangla text.

Table 10: Performance scores of fraud, partial fraud, micro avg., and weight avg. for corresponding five architectures.

Deep learning	Classification	Accuracy	Precision	Recall	F1-Score
LSTM	Fraud	71.15	72.73	64.00	68.09
	Partial Fraud		70.00	77.78	73.68
	Micro Avg.		71.36	70.89	70.88
	Weight Avg.		71.31	71.15	70.99
BiLSTM	Fraud	71.15	65.38	73.91	69.39
	Partial Fraud		76.92	68.97	72.73
	Micro Avg.		71.15	71.44	71.06
	Weight Avg.		71.82	71.15	71.25
CNN	Fraud	69.23	87.50	30.43	45.16
	Partial Fraud		63.64	96.55	76.71
	Micro Avg.		75.57	63.49	60.94
	Weight Avg.		74.19	67.31	62.76
CNN-BiLSTM	Fraud	73.08	67.86	82.61	74.51
	Partial Fraud		83.33	68.97	75.47
	Micro Avg.		75.60	75.79	74.99
	Weight Avg		76.49	75.00	75.05
CNN-LSTM	Fraud	75.00	84.62	74.83	61.11
	Partial Fraud		69.23	93.10	61.11
	Micro Avg.		76.92	70.46	70.26
	Weight Avg.		76.04	73.08	71.32

5 Results Discussion

There are various NLP libraries available for Bengali text mining including Keras and, extracting text data from Bangla is a highly challenging operation. It enabled us to create and assess deep learning models with improved functionality. A machine learning model uses vectorization to assist in extracting features from the data. We utilized five deep neural network models (CBF-SSL, CNN, LSTM, BiLSTM, CNN, and CNN-BiLSTM) in addition to seven machine learning algorithms (LR, RF, DT, KNN, Multi, NB, RBF, SVM, and SGD). We thoroughly examined a variety of performance metrics, including accuracy, recall, precision, F1-score, and support score, in order to assess the effectiveness of our classifier models. The SGD classifier algorithm demonstrated strong performance in machine learning, attaining an accuracy rate of 77.88%. On the other hand, the deep learning model for CBF-SSL (Customized BKash Fraud Semantic Sequential Learning) successfully answered 20% of the test cases with 75% accuracy. Table 11 projects that the supervised learning SGD is the

Table 11: Experimental results comparison using ML classifiers

Best accuracy	Most effective classifier	Least accuracy	Least effective classifier
77.88%	SGD	57.69%	RBF SVM

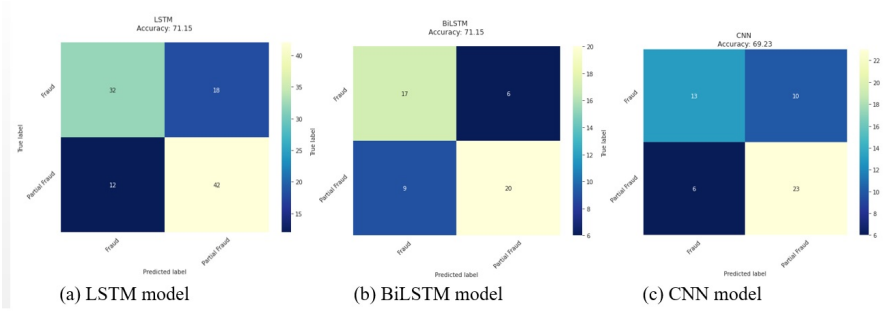
most effective classifier to classify the output significantly. On the other part, the experimental results for deep learning models are shown in Table 12, where we summarized the best performance models for deep learning models for input columns. For

Table 12: Experimental results comparison using DL architecture

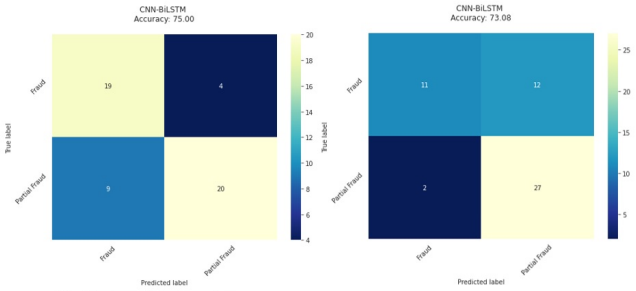
Best accuracy	Most effective classifier	Least accuracy	Least effective classifier
75%	CBF-SSL	67.31%	CNN

the evaluation purpose, we utilized a confusion matrix. In Figure 9 we presented performance evaluation on test data. A total of 400 fraud and non-fraud messages split into 80% and 20%. Here we give a result of 20%, which means, 52 text entries show the test evaluation. In our fraud identification efforts, 19 texts were correctly identified as fraudulent, and an additional 20 texts were partially identified as fraudulent. These results indicate that our model is capable of effectively detecting fraudulent messages.

Following the training and data validation processes, we visualized the accuracy of all deep learning methods, as depicted in Figure 10. The visualization of the CBF-SSL accuracy prediction reveals that the initial accuracy was below 0.40, eventually reaching a validation accuracy exceeding 0.80, while the training accuracy remained below 0.80. Conversely, our proposed model began with an initial accuracy of 0.40 and consistently achieved an accuracy of over 0.50 in both validation and training across each epoch later prediction accuracy was higher gradually. In contrast, the performance of other deep learning models declined over several epochs.

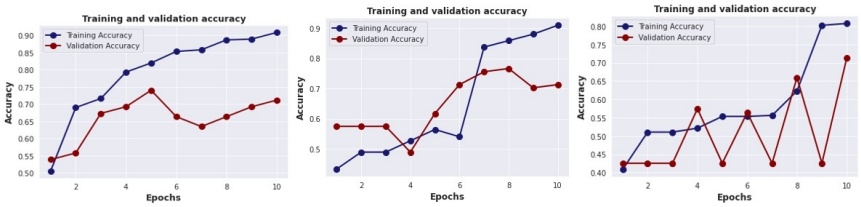


(a) LSTM model (b) BiLSTM model (c) CNN model

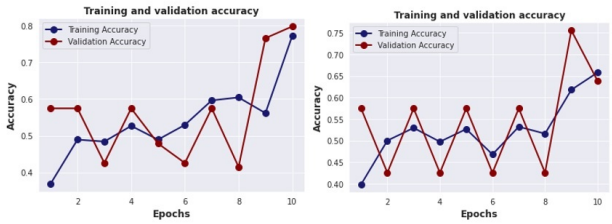


(d) CNN-LSTM model (e) CNN-BiLSTM model

Fig. 9: Demonstration of the confusion matrix for fraud and partial fraud text messages.(a) LSTM model (b) BiLSTM model (c) CNN model (d) CNN-LSTM model (e) CNN-BiLSTM model.



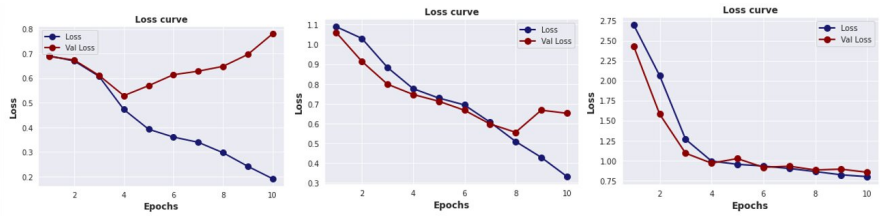
(a) Effectiveness of the LSTM (b) Effectiveness of the BiLSTM (c) Effectiveness of the CNN



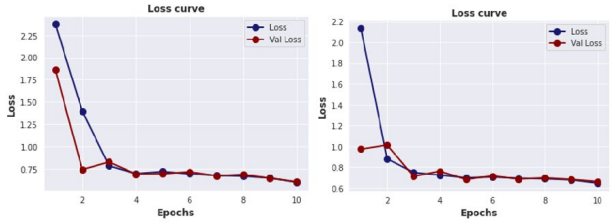
(d) Effectiveness of the CNN-LSTM network (e) Effectiveness of the CNN BiLSTM network

Fig. 10: Plot showing the relationship between each network's accuracy and epochs for training and validation.(a) LSTM (b) BiLSTM (c) CNN (d) CNN-LSTM, and (e) CNN-BiLSTM networks.

The loss graph serves as a crucial diagnostic tool in the realm of machine learning and deep learning. Its necessity lies in its ability to provide a visual representation of the performance of a model during the training process. By plotting the loss function against the number of training iterations or epochs, researchers and practitioners can observe how effectively the model is learning from the data. A decreasing loss indicates that the model is converging towards an optimal solution, while fluctuations or increasing loss may signify issues such as poor convergence, overfitting, or underfitting. Analyzing the loss graph allows for timely adjustments to the model architecture, learning rate, or regularization techniques, leading to more effective and robust models. Moreover, the loss graph facilitates communication and collaboration among team members by offering a clear and intuitive way to evaluate model performance and track progress throughout the training phase. Therefore, the loss graph is indispensable for understanding, optimizing, and validating machine learning models across various domains and applications.



(a) Loss of the LSTM (b) Loss of the BiLSTM (c) Loss of the CNN



(d) Loss of the CNN-LSTM network (e) Loss of the CNN BiLSTM network

Fig. 11: Plot showing the relationship between each network's loss curve for training and validation.(a) LSTM (b) BiLSTM (c) CNN (d) CNN-LSTM, and (e) CNN-BiLSTM networks.

In Figure 11, five distinct situations corresponding to epoch learning and loss rate are depicted. The loss function indicates that the LSTM validation loss is very high. Furthermore, in the case of the CNN model, both training and validation loss curves intersect between two lines, with a minor loss rate under 0.75. Lastly, in the graph representing CBF-SSL, the loss remains below 0.75 with a continuous epoch learning rate ranging from 2 to 20, showcasing stable and promising learning dynamics. The evaluation in Table 13 indicates that the CBF-SSL model has demonstrated superior

or comparable performance compared to the other models, as evidenced by the provided loss rates. Notably, the CBF-SSL model achieved the lowest mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), mean logarithmic error (MLSE), and root mean squared logarithmic error (RMLSE) values of 0.25, 0.25, 0.50, 0.1201, and 0.3466, respectively. Additionally, among the R2 values of all the models, the CBF-SSL model exhibited the highest R2 value of -0.0134, indicating a marginally better fit compared to the other models. These results underscore the effectiveness and robustness of the CBF-SSL model in accurately predicting outcomes and its potential as a valuable tool in the domain of analysis being conducted.

Table 13: Loss function rate for deep learning model

Loss function	Loss rate of deep learning models				
	LSTM	BiLSTM	CNN	CBF-SSL	CNN-BiLSTM
MAE	0.2884	0.2885	0.3077	0.25	0.2692
MSE	0.2885	0.2885	0.3076	0.25	0.2692
RMSE	0.5371	0.5371	0.5547	0.50	0.5188
R2	- 0.1556	- 0.1694	- 0.2474	-0.0134	- 0.0914
VARScore	- 0.1422	- 0.1559	- 0.2234	0.0201	0.0584
MSLE	0.1386	0.1385	0.1478	0.1201	0.1293
RMSLE	0.3722	0.3723	0.3844	0.3466	0.3596

6 Conclusion and Future Works

In this study, we employed two ensemble learning models, three deep learning models, and seven machine learning algorithms to detect complete and partial fraud in text messages. The dataset consisted of Bkash money-exchanging fraud messages obtained from the online resource Mendeley. Among the machine learning algorithms, the SVM algorithm demonstrated high accuracy during training. Nevertheless, utilizing a neural network with an ensemble learning model yielded an impressive performance of 75% accuracy, particularly with the CBF-SSL model. Performance evaluation scores further confirmed the accuracy of identifying text messages, even untrained data prior. The findings underscore the potential of advanced computational techniques in effectively identifying fraudulent activities within financial platforms. In this study, we use Bkash as a case study for being the most mature and acceptable financial services in Bangladesh. However, the same model would be relevant for other financial services in similar contexts and we contend that this matter is of paramount significance. Therefore, we argue that this piece of work is very timely and crucial for the Bangladeshi community, where the usage of financial services is increasing at a good pace among all sectors.

Additional research and experiments are required to enhance the accuracy and robustness of the models ensuring more reliable fraud detection. Moreover, inspecting the simulation results from Bkash text messages, we understand that it is crucial to examine the Bkash voice messages to identify the relevant fraud associated with voice

messages. Therefore, future works can be described in two-fold. Firstly, we would like to focus on speech-based fraud detection that will solve more precisely specific problems. Secondly, we intend to work from the output of our proposed research work to redesign the system model including text and speech messages with user feedback data.

Declarations

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- **Competing Interests:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Consent for Publication:** We affirm that this manuscript is unique, unpublished, and not under consideration for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors and that there are no other people who meet the criteria for authorship but are not listed. We further affirm that we have all approved the order of authors listed in the manuscript. We understand that the corresponding author is the sole contact for the editorial process. She is responsible for communicating with the other authors about progress, submissions of revisions, and final approval of proofs.
- **Data Availability:** Data will be made available on request.
- **Materials Availability:** Materials will be made available on request.
- **Code Availability:** This is private. But we can share the idea on request.

Credit authorship contribution statement

Sharun Akter Khushbu: Writing –review & editing, Conceptualization, Formal analysis, Data curation. **Fariha Tasmin Jaigirdar**: Formal analysis, Data curation, Writing –review & editing and Supervision. **Adnan Anwar**: Conceptualization, Methodology. **Ohidujjaman**: Writing –review & editing and Supervision.

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